

Talos

Spoken instructions for
learned dinner-table tasks

Two SO-101 arms
Six neural dinner skills
Bottle relays in both directions



Intel online challenge + Speechmatics integration

Six learned skills, one sequence

“Set the table”

Place the bottle, plate and mug.
Open the drawer.
Retrieve and place fork and spoon.

8/10 frozen full-sequence trials.
Each run starts a randomized scene.



Speech and control

Supported voice commands use Speechmatics

A human microphone trial passed on this PC.
The API key stays in the local server's .env file.

Bounded grammar and RGB scene checks choose the next skill.
Unsupported commands are refused. Progress and cancellation stay visible.

The hosted demo accepts typed commands. The local browser lab supports voice.

Learned bottle relay in both directions

Each arm releases onto the shared table

- 1 First arm lifts and releases the bottle on the table.
- 2 Both arms park before the second grasp.
- 3 Second arm places the bottle at its trained destination.

Frozen trials: 5/5 left to right and 5/5 right to left.

Two upright start regions and fixed goals. No airborne transfer.

How learned control works

Initial RGB features condition a small neural network

201 verified inputs train five new dinner skills.

The original bottle model stays intact. Four relay models use 52 inputs.

Joint feedback regulates neural motion.

A separate physical monitor checks contact, placement and parked arms.

Continuous visual correction remains unfinished.

The next experiment compares live and frozen camera input.

Physical evaluation

Frozen physical trials with every outcome retained

8 / 10

Six learned dinner tasks
Randomized task-preset starts

10 / 10

Bottle relay trials
Five in each direction

Dinner failures: mug 14.59 mm and spoon 13.61 mm XY error.
Fixed goals and small task-preset variations define this result.

Historical Intel bottle benchmarks

0.157 ms

Core i7-10850H CPU
FP32 neural inference

0.688 ms

Intel UHD integrated GPU
FP32 neural inference

These laptop results describe the original bottle model.
Final-suite Intel execution and Core Ultra eligibility remain open.

Network timing only. New models pass OpenVINO parity on this AMD/NVIDIA PC.
Core Ultra hardware execution has not been tested.

Reproducible baseline, active development

Code, models, compact inputs and complete outcomes

Six learned dinner skills complete without resets.

All frozen outcomes and exact training inputs are packaged.

The live demo is under private verification before release.

Still open: broad placement coverage and continuous visual correction.

Pouring, airborne handoffs and general object handling remain unfinished.

github.com/jannissio/talos-ai-infra